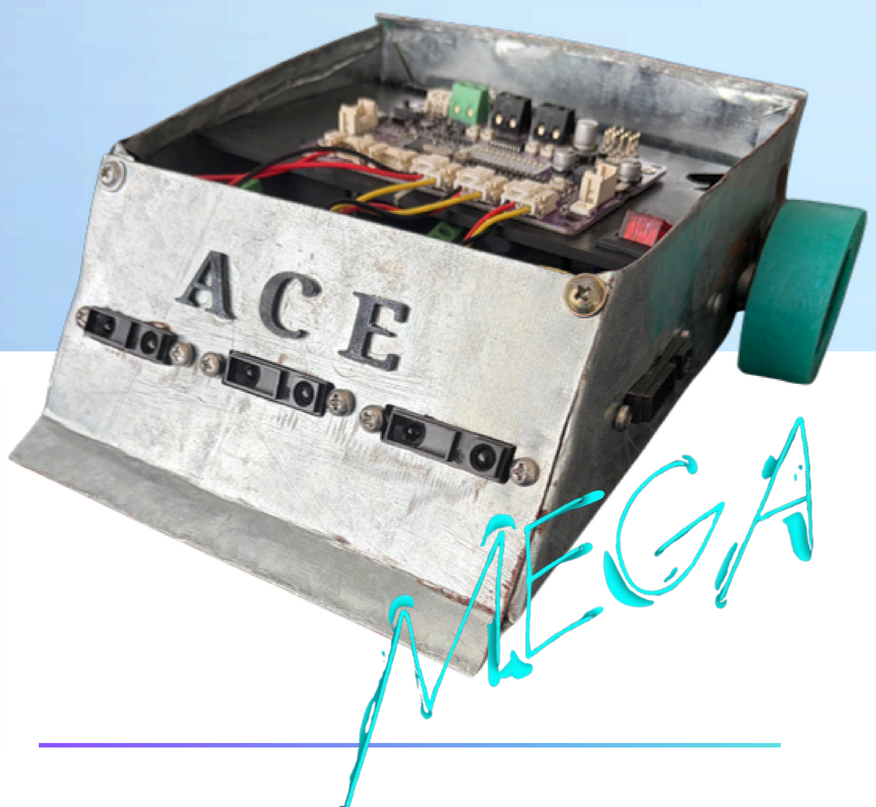
ACE MEGA

1kg Aluminum Chassis Sumobot



What is ACE Mega?

ACE Mega is the 1-kilogram sumobot in the ACE lineup. Initially, I planned to 3D-print the entire body and use an aluminum scoop. However, I decided to make it fully out of aluminum sheet instead. This was my first time working with an aluminum sheet chassis rather than relying on a 3D printer. The wheels were also custom-made—marking my very first attempt at DIY silicone wheels! **Fun fact:** I named it "Mega" because my favorite Transformer is Megatron!

Why ACE?

Same idea as ACE Micro: “keep acing,” the phrase I share with my juniors at Animo SPEAR. ACE is the name I’m building all my robots under.

I traced the layout on paper, then cut it with shears. I first planned a 3D-printed body with an aluminum scoop, but switched to full sheet metal to add weight and strength, and to experiment. The aluminum sheet came from our attic; my father loves DIY projects. Cutting, bending, and drilling were all hard, especially using a heavy-duty drill when I was used to handheld ones.

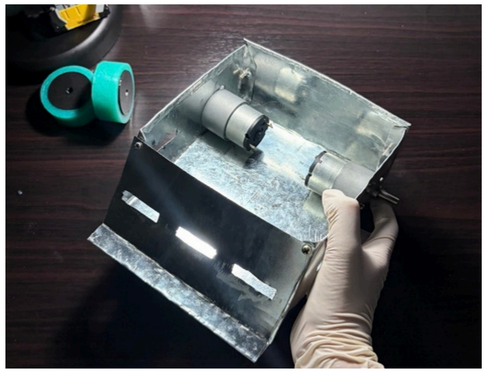


Figure 2. The chassis

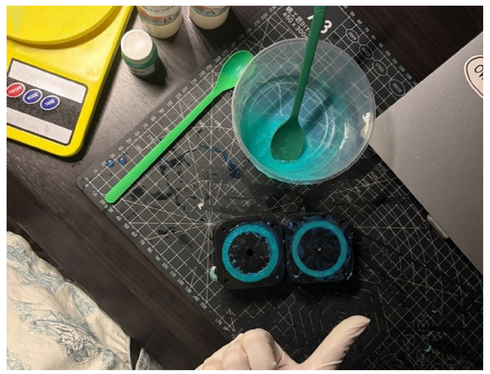
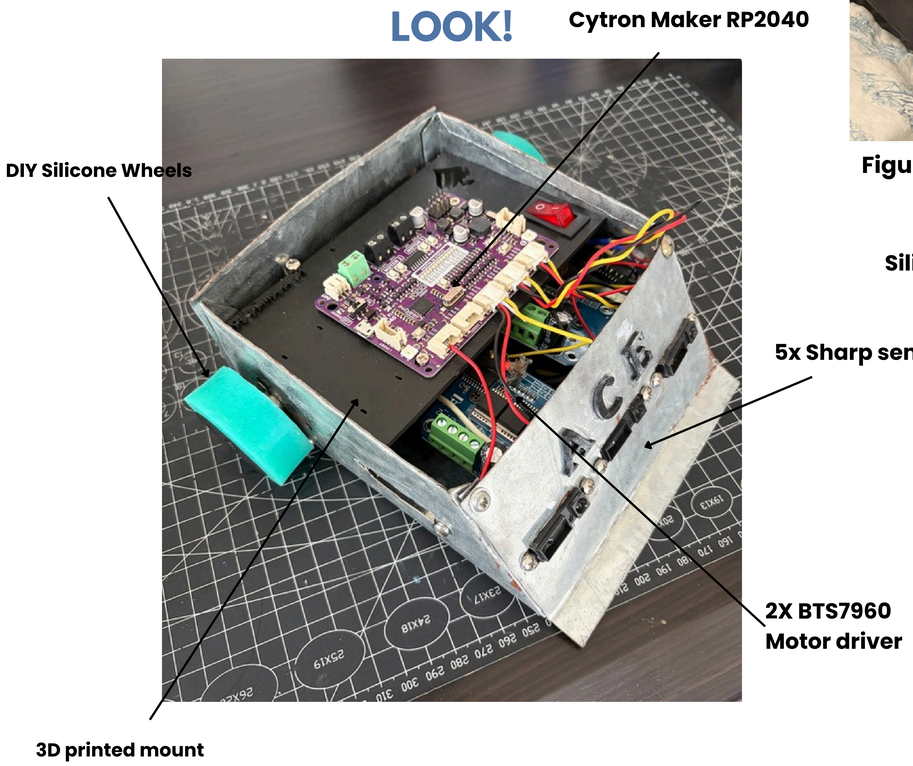


Figure 3. The creation of silicone wheels

This was my very first time making silicone wheels, and I used an exact 1:1 mixture ratio. They had bubbles and no traction at all, which I fixed in Micro. I made them blue because it is my favorite color—and the color I perceive best. **Fun fact:** I have minor protanopia, a form of color blindness!

LOOK!



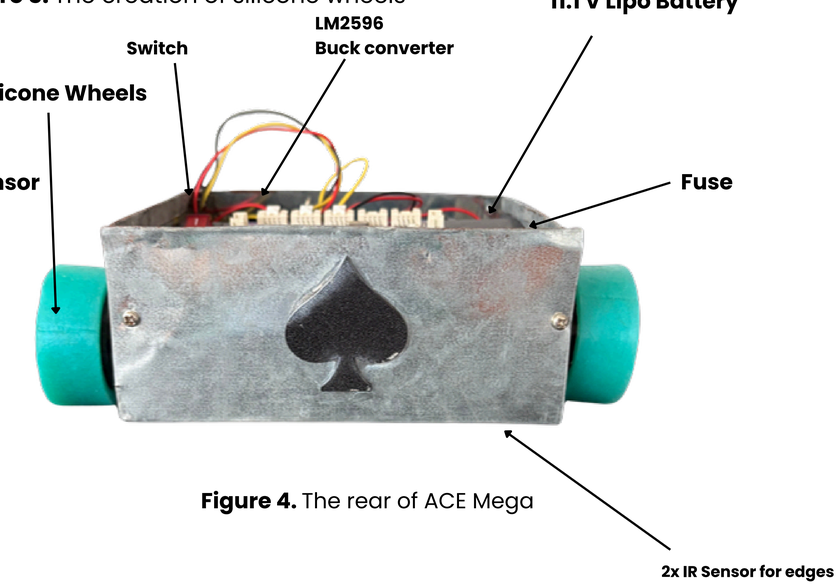


Figure 4. The rear of ACE Mega

The parts

ACE Mega features a 200x200 mm aluminum sheet chassis. Its main brain is a Cytron Maker RP2040, and two BTS7960 motor drivers control the motors. For perception, it uses five Sharp distance sensors to detect opponents and two IR line sensors to keep the sumobot from crossing the boundary of the dohyo ring. A LiPo of mine once smoked and its black cover ripped open, so now every robot gets a 15A automotive fuse on the battery line.

Where it all started

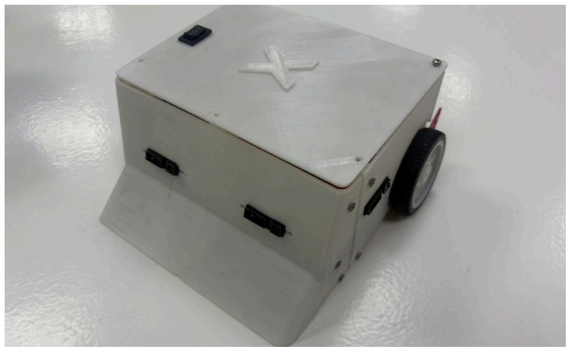


Figure 5. The Erovoutika Sumobot Kit

Same origin as ACE Micro. I started with line following, then got hooked on sumobots through the kit in our robolab and Kuya Zen.

ACE MEGA: THE PROCESS

	Expected	Outcome
Body	3D print body + aluminum wedge/scoop	Full aluminum
Drilling	Handheld drill	Heavy-duty drill
Wheels	Kit wheels	DIY Silicone wheels (1:1 mixture)
Battery	Standard Li-Po battery	Li-Po battery with 15A automotive fuse



Figure 1. Chassis layout marked on aluminum sheet

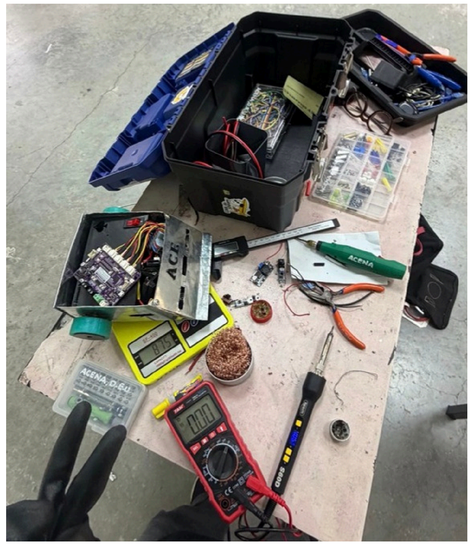


Figure 2. Workspace!

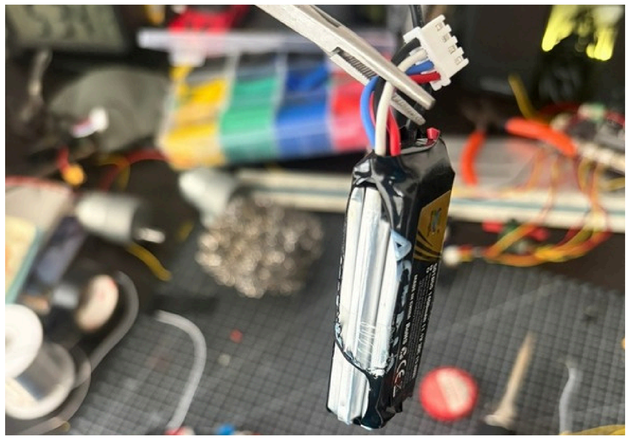


Figure 3. The LiPo that smoked that made me learn to put fuse next time. I added 15A automotive fuse.

What I learned: Sheet metal is much harder to work with than a 3D printer, but it gave me weight and strength. Every failure, from the wheels to the LiPo, changed how I built the next robot.